

FaceMe: An Augmented Reality Social Agent Game for Facilitating Children’s Learning about Emotional Expressions

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ABSTRACT

Children with autism spectrum disorder (ASD) experience dysfunctional emotional development leading to negative effects on their social communication. Although interventions are effective in helping children with ASD improve their social skills over time, they have been found to lack the essential ability to engage children in a real social environment. In this paper, we present “FaceMe,” which is an augmented reality (AR) system that uses a virtual agent and a set of tangible toolkits to teach children with ASD about six basic emotions and improve their emotional and communication skills. On the basis of the pilot data, the results suggest that children, especially those with ASD, were willing to socialize with the virtual agent and understand more emotional states. It is hoped that FaceMe can be used as a tool to provide assistance to children with ASD, as well as a way for future interface system design to support emotional development in children.

CCS CONCEPTS

• **Human-centered computing**; • **Human computer interaction (HCI)**; • **Interaction paradigms**; • **Mixed/augmented reality**;

KEYWORDS

Augmented reality, Social agent, Tangible user interface, Emotional development, Children

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1 INTRODUCTION

Children with autism spectrum disorder (ASD) have difficulties in managing their emotions and identifying basic facial expressions [1]. Conventional interventions use paper-based tools, such as cards, pictograms, and tangible objects, which are difficult and time-consuming to create, distribute, and use [2]. Computer-based tools have been shown to support children’s emotional states. These tools typically use two-dimensional (2D) graphics for animation and game design. However, the 2D environment cannot facilitate the children’s experience of a real social environment where they can learn to understand what emotions look like [3].

Recent research suggests that virtual agents are effective in supporting children’s emotional comprehension [4]. In this research study, we present “FaceMe” (Fig. 1(a)), an augmented reality (AR) social game that aims to help 4–7-year-old children with high-functioning ASD understand facial expressions. We designed an AR world that can overlay the real world with a three-dimensional (3D) virtual agent and facial expressions (Fig. 1(b) & (c)), with the aim of enriching children’s social experiences. To make the AR agent credible, we invited two teachers from a local autism child-care center to help guide the game design, including the agent’s appearance, speech, and gesture behaviors, in addition to the game mechanics. We designed a set of tangible toolkits to help the children manipulate the virtual information, to enhance their attention and engagement by bridging the boundary between the virtual and real worlds.

2 SYSTEM DESIGN

FaceMe is a single-player AR game using a 25-inch touchscreen that incorporates six types of emotions (happy, sad, angry, surprised, scared, and disgusted [5]) into three mini-games based on different learning goals: (1) Memory (Game 1), where the children must remember the virtual expressions on the screen and identify a certain expression after the agent flips and switches the order of the virtual facial expressions (Fig. 2(a)); (2) Animated Story Narratives (Game 2), where a story is played out on the touchscreen and the children must choose which expressions the game character should make during the story (Fig. 2(b)); and (3) Make Facial Expressions (Game 3), where the agent invites the children to make the expressions together by giving virtual hints (Fig. 2(c)). In Games 1 and 2, the child can control the 3D virtual expressions by moving the wooden markers on the table. Game 3 focuses on encouraging the children

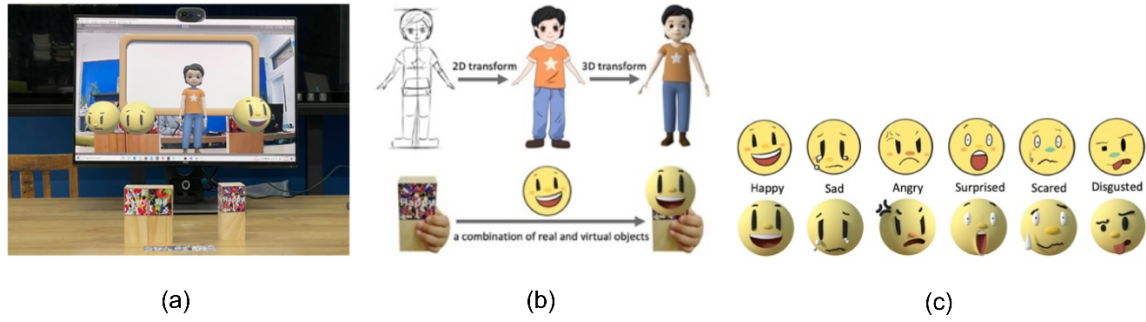


Figure 1: (a) An overview of the FaceMe system with its wooden markers, (b) the agent design, and (c) 3D facial expressions.

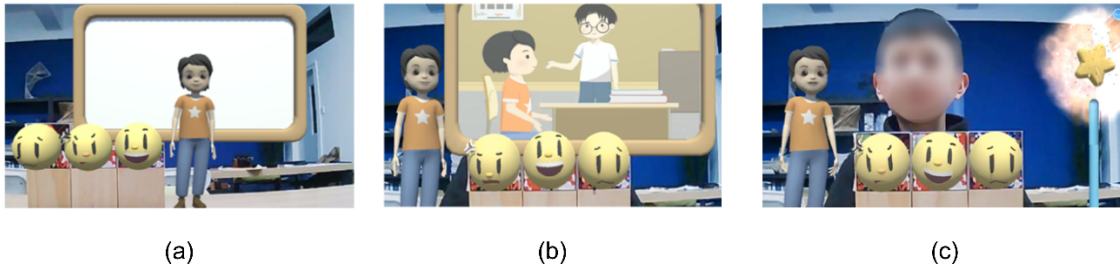


Figure 2: The mini-game interfaces for (a) Memory, (b) Animated Story Narratives, and (c) Make Facial Expressions.

to make facial expressions through matching with the facial recognition system to enhance their sense of engagement and judge whether the children have internalized the learning contents.

3 TECHNICAL IMPLEMENTATION

This system was developed using the Unity3D game engine, which consists of user and teacher sides. On the user side, the wooden markers are targeted by the Vuforia AR engine to generate 3D virtual expressions to interact with the children. The teacher side controls the real-time interaction between the agent and the children to trigger subsequent interactive events through the Android platform and UDP network protocol. In our system, a facial detection module was added based on the Facial Expression Recognition database (FER-2013) and trained using a convolution neural network [6]. When children make facial expressions, the information will be transmitted to the already-trained model for expression categorization and recognition.

4 EVALUATION

We recruited five children aged 4 to 7 years old (mean = 6.54, standard deviation = 0.71, 3 boys and 2 girls) to participate in this pilot study: three typically developed children (Pt1, Pt2, Pt3) and two children with high-functioning ASD (Pasd4, Pasd5). The studies involving human participants were reviewed and approved by the Ethical Review Board of Qingdao University. The parents provided their written informed consent for participation in this study. Our predominantly exploratory analyses of the pilot data tested the usability and engagement of the AR system. All children completed their tasks independently and four children felt that FaceMe was interesting. The delayed facial recognition in Game 3 (which was

a problem in most of the computer vision systems) made Pt2 feel that the virtual agent was “not smart.” During our observations, Pasd4 and Pasd5 showed a high level of engagement and fondness for the agent. Both children made several attempts to touch and hug the agent and Pasd4 was excited and stood up to make gestures to the agent. Pasd4 even gave the agent the name “Zhen Zhen,” which means “treasure” in the child’s local dialect. In particular, when the agent taught the child to recognize the expression “angry,” Pasd5 acted excited and scolded the agent for being rude, because to be “angry” was considered rude. We were pleased to discover that Pasd4 and Pasd5 treated the agent as a real partner and even increased their social behaviors with the agent beyond our expectations. The results suggest that FaceMe promoted reasoning about emotional states and increased communication and engagement to varying degrees, especially for Pasd4 and Pasd5.

5 CONCLUSION AND FUTURE WORK

In summary, these preliminary results show that FaceMe helped children to improve their engagement in facial expression learning, especially for two children with ASD. In our future works, we consider the laboratory environment an obstacle in social communication; therefore, we plan to use this system in actual classrooms that are familiar to the children with ASD to observe whether FaceMe can improve their emotional development and performance. Furthermore, we will conduct an iterative design process to further develop FaceMe by testing more subjects and including a night mode to support different user scenarios.

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